

An IPA-Kano model for classifying and diagnosing airport service attributes

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ABSTRACT

Considering the complexity of airport industry service offerings, it is important to identify which airport service attributes detract from passenger satisfaction and which attributes enhance it. Therefore, this study presents an IPA-Kano model to classify and diagnose the service attributes of an airport and suggests opportunities for service improvement. The model is validated through a case study of Taoyuan International Airport (TPE). The model combines the Kano model quality categories with the importance and performance of each service attribute and is developed for classifying and diagnosing service attributes at the TPE airport. As shown in this study, the proposed approach could help managers determine a correct and effective service improvement strategy for their airport. Another characteristic of the approach is its simplicity—advanced statistical knowledge is not required.

1. Introduction

An international airport creates passengers' first impressions of that country. Martín-Cejas (2006) recommends that airports provide remarkable services, and Kramer, Bothner, and Spiro (2013) also argue that providing an outstanding passenger experience has become a strategic priority for airports to increase their competitive advantage. Wattanacharoensil, Schuckert, Graham, and Dean (2017) argue that the airport "experience", which emerged in the early 2000s, has become a crucial factor in airport management for several reasons. One of the reasons is that a positive airport experience heightens passenger satisfaction, increases the likelihood of future visits by those individuals, and improves the reputation of the airport (Wattanacharoensil, Schuckert, & Graham, 2016). Therefore, understanding passenger perceptions and improving airport service quality are of paramount importance. According to the Airport Council International (ACI), airport service quality (ASQ) is the best way to heighten passenger satisfaction (Prentice & Kadan, 2019). ASQ is strongly correlated with passenger satisfaction (Abdul, Aziz, & Yahya, 2014) and is a critical driver of passenger satisfaction (Bogicevic, Yang, Bilgihan, & Bujisic, 2013). Therefore, airport managers should be committed to improving the service quality of an airport to increase passenger satisfaction.

Various approaches for evaluating and improving airport service quality are proposed. Parasuraman, Zeithaml, and Berry (1988) first developed the SERVQUAL model to evaluate service quality, which is based on a gap model. C-C Chou (2009) employed the SERVQUAL model to evaluate airport service quality. Similarly, Pabedinskaitė and

Akstinaitė (2014) evaluated the airport service quality by using the SERVQUAL model. The results indicate that the most important airport operational dimensions are closely related to the major services provided by the airlines. Tsai, Hsu, and Chou (2011) proposed a gap analysis model to assess the difference between passengers' perceptions and their expectations and to develop managerial strategies to improve airport service quality. Yeh and Kuo (2003) evaluated the service quality of some major airports in the Asia-Pacific area and proposed an overall service performance index that provides useful information to improve airport service in identified service areas. Fodness and Murray (2007) argued that airport service quality literature and research focus on the quality of attributes, which is distinguished from the gap theory model of the mainstream service quality perspective. Their study concluded that airport service quality has three key dimensions: function, interaction, and diversion. Mikulić and Prebežac (2008) proposed a three-step analytics approach, penalty-reward contrast analysis, impact range-performance analysis, and impact-asymmetry analysis, to develop a priority improvement strategy for airport service attributes. Liou, Tang, Yeh, and Tsai (2011) identified passenger perceptions using a rough set method and some situational decision rules in a preference model. Their results are useful to airport managers seeking to develop service improvement strategies. Bogicevic et al. (2013) performed a content analysis of traveller comments with a data-mining technique to identify satisfiers and dissatisfiers in the airport context. Chao, Lin, and Chen (2013) applied an importance–performance analysis (IPA) to evaluate priority improvement airport service attributes and to provide appropriate advice. Ching (2014) described satisfaction as a difference

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between passengers' expectations and their actual experience in the process of interacting with the airport. Kadaifci and Topcu (2014) proposed a decision support model based on fuzzy cognitive mapping to improve airport terminal service quality. Kurniawan, Sebbatu, and Davoudi (2017) measured the service quality of the SHIA airport using the SERVQUAL model, and the Kano model approach was used to conduct a conceptual development study to improve airport service quality that was oriented towards passenger perceptions. Airport studies indicate that service quality perceptions affect passengers' satisfaction (Bezerra & Gomes, 2015) (Mikulić & Prebežac, 2008). Consequently, service quality at all touchpoints influences the management strategies that airports implement as they seek to achieve customer satisfaction (Fernandes & Pacheco, 2010) (Pandey, 2016).

Due to the complexity of airport industry service offerings, it is crucial to determine which airport service attributes detract from customer satisfaction and which attributes enhance it. Generally, a linear relationship between attribute performance and customer satisfaction is assumed by the traditional techniques; thus, this assumption may engender wrong decisions on priority improvement strategies (Huiskonen & Pirttilä, 1998; Tontini & Silveira, 2007). Additionally, there are some service attributes that provide more satisfaction than others. Therefore, the key point is how to identify which service attribute can provide the most satisfaction. The Kano model (Kano, Hinterhuber, Bailon, & Sauerwein, 1984) is widely used to identify which quality attribute of a service provides much more satisfaction to customers. Additionally, it identifies which attributes do not engender satisfaction when present but cause dissatisfaction when they are not fulfilled. Understanding the service attributes in each category of the Kano model and developing appropriate actions can help to effectively improve service quality and sharpen customer satisfaction. The Kano model is used to categorize service attributes (customer requirements), but it neglects the performance and importance of those attributes. IPA, which was originated by Martilla and James (1977), is generally employed to identify the attributes that need to be improved. Consequently, this study introduces the IPA-Kano model for classifying and diagnosing service attributes to avoid categorization errors made by the IPA or the Kano model when either of these is used alone.

The present study combines the Skytrax scale with a proposed IPA-Kano model to classify service attributes and then diagnoses which service attributes should be prioritized or should be maintained with their current level of performance. The model is validated through a case study of Taoyuan International Airport (TPE). As shown in this study, the proposed approach could help determine a proper and effective service improvement strategy for the airport. Another characteristic of the approach is its simplicity—advanced statistical knowledge is not required.

2. Theoretical background

2.1. Skytrax scale

The ACI-ASQ and SKYTRAX scales are widely used as performance indicators for customer satisfaction that are applied to many airports (Kramer et al., 2013) (Paternoster, 2008). Since 2006, the ACI-ASQ has employed standard questionnaires with 34 performance metrics reflecting passengers' perceptions of airport service attributes. Skytrax agency also uses a ranking system for passenger satisfaction surveys according to 39 airport service attributes, called the Skytrax scale in this study, as shown in Table 1. These attributes include key attributes of process efficiency, terminal comfort and cleanliness, passenger facilities and staff service.

The current study used the Skytrax scale for three reasons. First, the ACI/ASQ scale has been widely used in many studies (Bezerra & Gomes, 2016) (Bogicevic et al., 2013; Chien-Chang Chou, Liu, Huang, Yih, & Han, 2011) (Fernandes & Pacheco, 2010) (Lupo, 2015) (Martín-Domingo, Martín, & Mandsberg, 2019). This study attempts to adopt

the Skytrax scale, which is a different but still widely used scale by many airports. Second, as part of Skytrax, AirlineQuality.com is the largest airline and airport review community that includes reviews of 682 airline companies and 725 airports worldwide (Bogicevic, Yang, Bujisic, & Bilgihan, 2017). The Skytrax scale began taking reviews from passengers on its website in 2010. The third reason is that Skytrax's airport rankings and "World Airport Awards" attract international interest and are used for promotional purposes by some airports (Pérezgonzález & Gilbey, 2011).

According to (Skytrax, 2019), the world's top five airports for 2019 were Singapore Changi, Tokyo Haneda, Seoul Incheon, Doha Hamad, and Hong Kong. Four out of the top five airports are located in Asia. The TPE airport ranked 13, far behind other neighbouring Asian airports, which implies that the service quality delivered by the TPE airport is not in line with passengers' expectations.

2.2. Kano two-dimensional model

Traditionally, customer satisfaction has been mostly treated in a one-dimensional way: the higher the perceived quality of a product or service is, the higher the customer's satisfaction is and vice versa. Fulfilling individual product or service requirements to a certain extent does not necessarily engender a higher level of customer satisfaction. Consequently, Professor Kano and his colleague introduced a two-dimensional construction, termed the Kano model, that considers the asymmetric and nonlinear relationship between product or service performance and customer satisfaction. The Kano methodology evaluates the characteristics of product or service requirements by designing a dysfunctional and functional questionnaire to examine customer satisfaction with product or service quality attributes and then classifies them into one of the following five quality categories.

1. Attractive (A): Fulfilling these quality attributes leads to an increase of more than proportional satisfaction. If not fulfilled, they do not cause dissatisfaction. Attractive attributes are difficult to identify because they are latent and not expected by the customer.
2. One-dimensional (O): Customer satisfaction is proportional to the extent of fulfilling these attributes and vice versa. One-dimensional attributes are usually clearly expressed by the customer.
3. Must-be (M): The customer will be highly dissatisfied if these attributes are not fulfilled. Must-be attributes are basic criteria of a product or service and a certain minimal quality level for the product or service must be reached by the company.
4. Indifferent (I): The customer does not care about whether these attributes are fulfilled. Consequently, indifferent attributes are those whose presence or absence does not lead to customer satisfaction or dissatisfaction.
5. Reverse (R): The customer does not want these requirements or attributes, and if the product or service meets them, the customer will be dissatisfied.

In the lifecycle of a product or service, quality attributes start in the indifferent category, go to attractive, then one-dimensional, and end as must-be. This means that a quality attribute that is deemed attractive now may become a must-be attribute later (Noriaki, 2001). Robertshaw (1995) suggested that the priority should be to first fulfil the must-be attributes; the next priority is fulfilling the one-dimensional attributes, and the final priority is fulfilling the attractive attributes.

2.3. Importance-performance analysis

Martilla and James (1977) originated importance-performance analysis (IPA). IPA is a technique that can identify attributes for improvement based on user evaluation of the importance and performance of each product or service attribute (Sampson & Showalter, 1999). IPA is a simple but effective tool that can help managers

Table 1
Skytrax scale of airport service attributes.

No.	Attribute description	No.	Attribute description
1	Getting to and from the airport, ease of access	21	Cleanliness of washroom facilities
2	Public transport options, efficiency and prices	22	TV and entertainment facilities
3	Taxi availability and prices	23	Quiet areas, day rooms, hotel facility, rest areas
4	Availability of luggage trolleys (airside & landside)	24	Children's play area and facilities provided
5	Terminal comfort, ambience and design	25	Choice of shopping - tax free and other outlets
6	Terminal cleanliness, floors, seating and public areas	26	Prices charged in retail outlets
7	Seating facilities throughout terminals	27	Choice of bars, cafes and restaurants
8	Immigration - queuing times/system	28	Prices charged in bars, cafes and restaurants
9	Immigration - staff attitude	29	Wi-Fi service
10	Waiting times at security screening	30	Power charging facilities
11	Courtesy and attitude of security staff	31	Telephone and fax locations
12	Check-in facilities, queuing systems and seating	32	Bureau de change facilities
13	Wayfinding and terminal signage	33	ATM facilities
14	Clarity of boarding calls and airport PAs	34	Smoking policy/smoking lounges
15	Flight info screens - clarity/quality of information	35	Standards for disabled user access and facilities
16	Friendliness of airport staff	36	Baggage delivery times
17	Language skills for airport staff	37	Priority baggage delivery efficiency
18	Ease of transit through airport	38	Lost luggage services
19	Location of airline lounges	39	Perception of security and safety standards
20	Washroom and shower facilities in terminal		

Table 2
Series, categories, and strategic priorities of the IPA-Kano model.

Strategic series	Kano	Importance	Performance	IPA strategy	IPA-Kano category	Strategic priority	
						Improvement	Keep up the good work
Hygiene series	M	High	High	Keep up the good work	Survival		1
	M	High	Low	Concentrate here	Fatal	1	
	M	Low	Low	Low priority	Chronic disease	2	
	M	Low	High	Possible overkill	Fitness		2
War series	O	High	High	Keep up the good work	Major weapon		3
	O	High	Low	Concentrate here	Defenceless strategic point	3	
	O	Low	Low	Low priority	Defenceless zone	4	
	O	Low	High	Possible overkill	Supportive weapon		4
Treasure series	A	High	High	Keep up the good work	Precious treasure		5
	A	High	Low	Concentrate here	Dusty diamond	5	
	A	Low	Low	Low priority	Rough stone	6	
	A	Low	High	Possible overkill	Beginning jewellery		6

effectively utilize limited resources in product or service quality management to achieve maximum customer satisfaction. In IPA, two dimensions of performance and importance of each attribute are analysed. These two dimensions can then be combined into a matrix that identifies to managers the attributes that should be improved. The importance and performance pairs for each attribute are plotted in the matrix. The horizontal axis represents the degree of importance, and the vertical axis denotes the degree of performance. By using mean scores as the crosshairs, attributes can be assigned to one of the four quadrants. Attributes with high importance and high performance are categorised in Quadrant I, which is labelled 'Keep up the good work'. Attributes in this quadrant are considered very important, and the performance for these attributes should be maintained at a high level. Attributes with low importance and high performance are categorised in Quadrant II, which is labelled 'Possible overkill'. Attributes in this quadrant require excessive resources that should be reallocated to address more important attributes. Attributes with low importance and low performance are categorised in Quadrant III, which is tagged as 'Low priority'. Attributes in this quadrant have a low priority for improvement. Attributes with high importance and low performance are categorised in Quadrant IV, which is labelled 'Concentrate here'. Attributes in this quadrant should receive the highest priority for improvement.

3. Methodology

The present study combines the Skytrax scale with the IPA-Kano model, originated by Kuo, Chen, and Deng (2012), to classify service attributes and then diagnose which service attributes should be prioritized for attention or should be maintained at their current level of performance. The IPA-Kano model considers three Kano categories of service attributes (must-be, one-dimensional, and attractive), two levels of attribute importance (high and low), and two levels of attribute performance (high and low) and classifies service attributes into three series and twelve categories. Using the IPA-Kano model, airport managers can comprehensively understand passenger perceptions of all service attributes and develop corresponding strategies to address each one. In this study, the three series and twelve categories are named using familiar terms. The three series are named hygiene, war, and treasure, which correspond to must-be, one-dimensional, and attractive, respectively. Must-be attributes, classified in the hygiene series, are as basic to firms as health is to the human body. Underperformance of these attributes will cause customers to have the highest level of dissatisfaction. One-dimensional attributes, classified in the war series, are a basis for competition among airports in the industry 'battlefield'. Passenger satisfaction is proportionally increased with the level of sufficiency, and thus, these attributes are viewed as the major points over which companies fight with opponents. Attractive attributes, classified in the treasure series, provide passengers with unexpected feelings of pleasure and lead to a high level of satisfaction.

Table 3
Dimensions and attributes of TAISERQ (Tseng & Lin, 2018).

Attributes	Ambience (AM)	Convenience (CO)	Personnel (PE)	Empathy (EM)	Mobility (MO)	Baggage (BA)
Children's play area and facilities provided	0.750					
Smoking policy/smoking lounges	0.742					
Washroom and shower facilities in terminal	0.718					
Cleanliness of washroom facilities	0.670					
Terminal comfort, ambience and design	0.660					
Location of airline lounges	0.624					
Quiet areas, day rooms, hotel facility, rest areas	0.616					
Terminal cleanliness, floors, seating and public areas	0.609					
Seating facilities throughout terminals	0.599					
TV and Entertainment facilities	0.509					
Getting to and from the airport, ease of access		0.766				
Choice of bars, cafes and restaurants		0.724				
Prices charged in retail outlets		0.701				
Standards of disabled user access and facilities		0.665				
Prices charged in bars, cafes and restaurants		0.663				
Availability of luggage trolleys (airside & landside)		0.653				
Public transport options, efficiency and prices		0.582				
Choice of shopping - tax free and other outlets		0.509				
Check-in facilities, queuing systems and seating			0.824			
Language skills for airport staff			0.788			
Immigration - queuing times/system			0.759			
Immigration - staff attitude			0.753			
Friendliness of airport staff			0.675			
Courtesy and attitude of security staff			0.576			
Waiting times at security screening			0.571			
ATM facilities				0.635		
Power charging facilities				0.565		
Wi-Fi service				0.534		
Telephone and fax locations				0.530		
Bureau de change facilities				0.527		
Way finding and terminal signage					0.862	
Clarity of boarding calls and airport PAs					0.743	
Perception of security and safety standards					0.714	
Ease of transit through airport					0.648	
Baggage delivery times						0.587
Priority baggage delivery efficiency						0.541
Lost luggage services						0.529
Reliability coefficient (alphas)	0.884	0.867	0.837	0.830	0.788	0.780
Eigenvalue	4.282	3.325	2.388	2.230	1.965	1.632
Percent of variance explained	18.600	14.986	11.455	10.026	8.311	6.411
Cumulative percent of variance explained	18.600	33.586	45.041	55.067	63.378	69.789

Table 4
Sample demographic data.

Item	Description	IPA		Kano	
		Freq.	%	Freq.	%
Gender	Male	242	36.80%	236	37%
	Female	416	63.20%	401	63%
Age	20–30	350	53.20%	314	49.30%
	31–40	273	41.50%	271	42.50%
	41–50	27	4.10%	42	6.60%
	51–60	6	0.90%	10	1.60%
	61+	2	0.30%	0	0%
Nationality	Taiwan	658	100%	630	98.90%
Times used in previous year	Once	195	29.60%	174	27.30%
	twice	233	35.40%	222	34.90%
	thrice or more	230	35.00%	241	37.80%

Unfulfillment of attributes in this series does not engender passenger dissatisfaction, but fulfilment of these attributes greatly increases customer satisfaction. Such attributes resemble treasure: people can live without treasure but are excited when they obtain it.

Each series is further classified into four categories by four IPA quadrants. The four categories of the hygiene series are survival (must-be, high performance and high importance), fatal (must-be, low performance and high importance), chronic disease (must-be, low performance and low importance), and fitness (must-be, high performance and low importance). Survival attributes do not engender passenger

dissatisfaction. However, dissatisfaction immediately arises once the attribute performance declines. Hence, preserving these attributes with higher performance is a key to firm survival. Fatal attributes will cause a high degree of passenger dissatisfaction. For managers, improvement of these attributes should be given the highest priority. For chronic disease attributes, lower importance means that passengers may overlook poor attribute performance initially. Similar to chronic diseases in the human body, these attributes do not damage firms immediately. Thus, for managers, the chronic disease attributes should be the priority to address after the fatal attributes. Fitness attributes are less important than survival attributes. They do not influence firm survival but should still be maintained at a certain performance level.

The four categories of the war series are major weapon (one-dimensional, high performance and high importance), defenceless strategy point (one-dimensional, low performance and high importance), defenceless zone (one-dimensional, low performance and low importance), and supportive weapon (one-dimensional, high performance and low importance). Major weapon attributes can be used to help managers repel competitors. Managers should always make efforts to keep the attribute at a higher level of performance to repel the rivals. Defenceless strategy point attributes are similar to a completely unguarded fortified point during wartime. Once under attack, the lack of defence can be fatal. These attributes may also be deemed 'must-be' attributes and should be ranked as the third priority for improvements. Defenceless zone attributes are similar to a space that needs protection but is not guarded at all during wartime. Such a category is not as

Table 5
Kano model classification of service attributes of the TAISERQ scale.

ID	O	M	A	I	R	Q	Category
AM1	9.42%	6.75%	28.10%	52.12%	3.30%	0.31%	I
AM2	10.05%	47.88%	7.06%	35.01%	0.00%	0.00%	M
AM3	43.96%	41.13%	9.58%	5.34%	0.00%	0.00%	O
AM4	46.47%	43.64%	5.49%	4.40%	0.00%	0.00%	O
AM5	25.27%	50.24%	9.42%	15.07%	0.00%	0.00%	M
AM6	10.83%	10.36%	32.18%	45.37%	1.26%	0.00%	I
AM7	26.37%	13.50%	40.03%	19.78%	0.31%	0.00%	A
AM8	28.26%	51.18%	13.34%	7.22%	0.00%	0.00%	M
AM9	17.43%	18.37%	33.91%	30.30%	0.00%	0.00%	A
AM10	9.42%	8.01%	33.75%	48.04%	0.78%	0.00%	I
CO1	41.60%	39.09%	11.30%	8.01%	0.00%	0.00%	O
CO2	13.34%	15.23%	37.99%	30.30%	3.14%	0.00%	A
CO3	26.22%	30.93%	19.94%	22.92%	0.00%	0.00%	M
CO4	36.11%	20.09%	27.94%	15.54%	0.31%	0.00%	O
CO5	11.15%	40.66%	10.99%	36.89%	0.31%	0.00%	M
CO6	25.27%	51.65%	8.01%	15.07%	0.00%	0.00%	M
CO7	38.93%	33.44%	18.21%	9.42%	0.00%	0.00%	O
CO8	35.79%	26.84%	24.65%	12.72%	0.00%	0.00%	O
PE1	31.08%	49.76%	9.26%	9.89%	0.00%	0.00%	M
PE2	41.92%	24.49%	24.33%	9.26%	0.00%	0.00%	O
PE3	25.59%	53.22%	8.48%	12.72%	0.00%	0.00%	M
PE4	39.09%	26.37%	24.65%	9.89%	0.00%	0.00%	O
PE5	36.73%	24.49%	24.33%	14.44%	0.00%	0.00%	O
PE6	33.91%	25.59%	28.26%	12.24%	0.00%	0.00%	O
PE7	37.68%	32.97%	16.01%	13.34%	0.00%	0.00%	O
EM1	27.16%	35.16%	21.51%	16.17%	0.00%	0.00%	M
EM2	31.24%	10.36%	43.17%	15.23%	0.00%	0.00%	A
EM3	34.22%	20.09%	37.21%	8.01%	0.47%	0.00%	A
EM4	18.37%	36.73%	11.30%	32.65%	0.94%	0.00%	M
EM5	24.65%	47.72%	24.18%	3.45%	0.00%	0.00%	M
MO1	20.09%	59.18%	1.88%	17.90%	0.94%	0.00%	M
MO 2	27.00%	49.29%	19.00%	4.71%	0.00%	0.00%	M
MO 3	19.00%	59.18%	10.83%	10.99%	0.00%	0.00%	M
MO 4	25.27%	61.22%	8.95%	4.55%	0.00%	0.00%	M
BA1	25.43%	47.57%	10.83%	16.17%	0.00%	0.00%	M
BA2	15.86%	14.76%	37.52%	31.87%	0.00%	0.00%	A
BA3	22.76%	29.67%	41.92%	5.65%	0.00%	0.00%	A

Notes: A = Attractive; O = One-Dimensional; M = Must-Be; I = Indifferent; R = Reverse; Q = Questionable.

important as a defenceless strategy point but remains a weak spot, and thus improvement is required. Consequently, these attributes are ranked as the fourth priority for improvement. Supportive weapon attributes have similar characteristics to major weapons and can be treated as supportive weapons. It is strongly suggested to turn supportive weapons into major weapons when enough resources are available.

Finally, the four categories of the treasure series are precious treasure (attractive, high performance and high importance), dusty diamond (attractive, low performance and high importance), rough stone (attractive, low performance and low importance), and beginning jewellery (attractive, high performance and low importance). Precious treasure attributes are enchanting to passengers and thus are the precious treasure of firms. Managers should promote these attributes and make them more noticeable to passengers. Dusty diamond attributes resemble diamonds covered with dust that need to be wiped clean so they will glitter. If enough resources are available, managers are urged to improve these attributes to attract customers. Rough stone attributes with low importance and low performance resemble rough stones that need to be carved. If enough resources are available, improving all such low-performance attributes is strongly recommended. Beginning jewellery attributes possess low importance and high performance, like real-world beginning jewellery. If enough resources are available, managers can use some marketing approaches to enhance the perceived importance of these attributes and change them into precious treasures.

The appropriate coping strategies for each service attribute analysed by the IPA and IPA-Kano models are summarized in Table 2.

3.1. Factor analysis and reliability analysis

Tseng and Lin (2018) performed factor analysis to reduce 39 service attributes of the original Skytrax scale to an interpretable and manageable set of factors based on attribute performance values. In Tseng and Lin's study, they used principal component analysis with varimax rotation and generated six dimensions and 37 service attributes, called the Taiwan Airport Service Quality (TAISERQ) attribute scale, which is shown in Table 3, where eigenvalues were set at greater than 1.0. Six dimensions, which can be summarized as Ambience (0.88), Convenience (0.84), Personnel (0.87), Empathy (0.83), Mobility (0.79), and Baggage (0.78), cumulatively accounted for 69.79% of the variance. Table 3 shows the factor loadings after rotation (only those greater than 0.50 are displayed). The values in parentheses represent Cronbach's reliability coefficients, ranging from 0.78 to 0.88, which indicate considerable reliability (Nunnally, 1978) (Drost, 2011).

3.2. Questionnaire design

The questionnaires used in this study consist of three parts, based on the TAISERQ scale. The first part is the Kano questionnaire, which includes pairs of functional and dysfunctional questions addressing the attributes of each passenger. One question concerns the passenger response if the TPE airport fulfils the attribute (i.e., the functional form of the question); the other concerns the passenger response if the TPE airport lacks the attribute (i.e., the dysfunctional form of the question). In assessing a service attribute, survey participants choose one of five options (Matzler & Hinterhuber, 1998): (1) I dislike it that way; (2) I can live with it that way; (3) I am neutral; (4) It must be that way; (5) I like it that way. The second part is the IPA questionnaire evaluating passenger perceptions of attribute importance and performance as delivered by the TPE. Each importance/performance item is rated using a five-point Likert scale, where "1 = highly unimportant/strongly dissatisfied", "2 = not important/dissatisfied", "3 = neutral/neutral", "4 = important/satisfied", and "5 = very important/strongly satisfied". The third part requests the basic personal data of the respondents, including the frequency of their use of TPE over the past year.

The target population for this study is passengers who have travelled from or to TPE. The sample survey uses convenience sampling by requesting passengers who are waiting at departure lounges and arrival lounges in the TPE airport terminal to complete a web-based questionnaire (by scanning a QR code to link to the questionnaire) or a paper-based questionnaire. The survey was conducted from 1 August to 30 September 2019, and a total of 700 samples were collected, of which 657 IPA and 637 Kano samples were verified as useful.

3.3. Sample profile

Table 4 presents the demographic information of the sample. Approximately 63.20% of the respondents of IPA and Kano questionnaires were female. In terms of age, approximately 84.7% of respondents of the IPA and 91.8% of the Kano questionnaires were under 40 years old. Over 70% of respondents had used TPE at least twice or more from 1 September 2017 to 30 September 2019. The sample seems heavily skewed towards female and the youngers.

4. Results and discussion

4.1. Kano model evaluation

The frequencies of responses to the Kano model classification evaluations are investigated to categorize each attribute of the TAISERQ scale. Table 5 shows the categorised results. The frequencies of the TAISERQ scale attributes labelled must-be (M), one-dimensional (O), attractive (A), and indifference (I) are found to be 16, 11, 7, and 3, respectively. Nearly half (43.2%) of the attributes are in the must-be

Table 6
Attribute importance, performance, and GAP of the TAISERQ scale.

Dimension	ID	Attribute	Importance TPE (A)	Performance TPE (B)	GAP Paired (B-A)	p-value
Ambience	AM1	Children's play area and facilities provided	3.26(L)	3.27(L)	0.01	0.16
	AM2	Smoking policy/smoking lounges	3.94(L)	3.48(L)	-0.48	0.00
	AM3	Washroom and shower facilities in terminal	4.70(H)	3.89(H)	-0.83	0.00
	AM4	Cleanliness of washroom facilities	4.75(H)	3.92(H)	-0.81	0.00
	AM5	Terminal comfort, ambience and design	4.32(H)	3.55(L)	-0.76	0.00
	AM6	Location of airline lounges	3.44(L)	3.38(L)	-0.09	0.01
	AM7	Quiet areas, day rooms, hotel facility, rest areas	4.00(L)	3.34(L)	-0.70	0.00
	AM8	Terminal cleanliness, floors, seating and public areas	4.81(H)	3.78(H)	-1.04	0.00
	AM9	Seating facilities throughout terminals	4.34(H)	3.62(H)	-0.75	0.00
	AM10	TV and entertainment facilities	3.09(L)	3.36(L)	0.21	0.00
Convenience	CO1	Getting to and from the airport, ease of access	4.89(H)	3.80(H)	-1.06	0.00
	CO2	Choice of bars, cafes and restaurants	4.15(L)	3.07(L)	-1.16	0.00
	CO3	Prices charged in retail outlets	4.09(L)	3.21(L)	-0.95	0.00
	CO4	Standards of disabled user access and facilities	4.49(H)	3.72(H)	-0.79	0.00
	CO5	Prices charged in bars, cafes and restaurants	4.20(L)	2.96(L)	-1.39	0.00
	CO6	Availability of luggage trolleys (airside and landside)	4.11(L)	3.89(H)	-0.20	0.00
	CO7	Public transport options, efficiency and prices	4.60(H)	3.73(H)	-0.86	0.00
	CO8	Choice of shopping - tax free and other outlets	3.84(L)	3.53(L)	-0.38	0.00
Personnel	PE1	Check-in facilities, queuing systems and seating	4.53(H)	3.65(H)	-0.91	0.00
	PE2	Language skills for airport staff	4.47(H)	4.00(H)	-0.47	0.00
	PE3	Immigration - queuing times/system	4.80(H)	3.56(L)	-1.28	0.00
	PE4	Immigration - staff attitude	4.19(L)	3.88(H)	-0.30	0.00
	PE5	Friendliness of airport staff	4.28(H)	3.96(H)	-0.29	0.00
	PE6	Courtesy and attitude of security staff	3.97(L)	3.84(H)	-0.12	0.00
	PE7	Waiting times at security screening	4.64(H)	3.80(H)	-0.84	0.00
Empathy	EM1	ATM facilities	4.08(L)	3.57(L)	-0.51	0.00
	EM2	Power charging facilities	4.34(H)	3.56(L)	-0.81	0.00
	EM3	Wi-Fi service	4.41(H)	3.69(H)	-0.72	0.00
	EM4	Telephone and fax locations	3.24(L)	3.22(L)	-0.12	0.00
	EM5	Bureau de change facilities	4.00(L)	3.66(H)	-0.36	0.00
Mobility	MO1	Way finding and terminal signage	4.78(H)	3.84(H)	-0.92	0.00
	MO 2	Clarity of boarding calls and airport PAs	4.63(H)	3.71(H)	-0.93	0.00
	MO 3	Perception of security and safety standards	4.65(H)	3.71(H)	-0.93	0.00
	MO 4	Ease of transit through airport	4.66(H)	3.76(H)	-0.87	0.00
Baggage	BA1	Baggage delivery times	4.63(H)	3.21(L)	-1.51	0.00
	BA2	Priority baggage delivery efficiency	4.01(L)	3.39(L)	-0.68	0.00
	BA3	Lost luggage services	4.69(H)	3.41(L)	-1.29	0.00
		grand mean =	4.27	3.59		

Note: (L) means Low importance/performance; (H) means High importance/performance

category, 29.7% are one-dimensional, 18.9% are attractive, and 0.82% are indifferent.

4.2. GAP score analysis

The GAP score of an attribute is the difference between the attribute performance of TPE and its corresponding attribute importance. Subsequently, paired sample *t*-tests are employed to examine whether there is a significant difference between the attribute performance of TPE and its corresponding attribute importance. Table 6 shows that all attributes except AM1 have significantly negative GAP for the TPE airport, indicating that most of the passengers have higher expectations (importance) than perceptions (performance) of the airport services provided by TPE.

4.3. IPA results

A total of 30 out of the 37 service attributes have mean values of attribute importance above 4.0 (refer to Table 6). Thus, one might assume that respondents have high expectations. The results of the IPA approach are shown in columns 4–5 of Table 6 and are also plotted in Fig. 1. Crosshairs (horizontal and vertical lines) are set as their gross mean values, and a third line at a 45-degree angle where performance equals importance is also plotted. The cut-off point of 'High' versus 'Low' importance for all service attributes is 4.27, and the cut-off point of 'High' versus 'Low' performance is 3.59. As seen in Fig. 1, only one attribute, 'TV and entertainment facilities (AM10)', is above the 45-degree line, representing that only AM10's performance score (= 3.66)

is superior to its importance score (= 3.09). A total of eleven attributes out of 37 attributes fell into the 'low priority' category, indicating that these attributes' importance and their perceived performance quadrant are both low. Thus, they have low priority. In total, sixteen service attributes, including AM3, AM4, AM8, AM9, CO1, CO4, CO7, PE1, PE2, PE5, PE7, EM3, MO1, MO2, MO3, and MO4, are positioned in the 'keep up the good work' quadrant due to their higher values of importance and performance. It can be considered that these attributes are the strengths of the TPE airport. Finally, the 'Concentrate here' quadrant contains five attributes: 'Terminal comfort, ambience and design (AM5)', 'Immigration - queuing times/system (PE3)', 'Power charging facilities (EM2)', 'Baggage delivery times (BA1)', and 'Lost luggage services (BA3)'. Passengers seem to think that the TPE does not perform these attributes well, but the attributes are important to them. For this reason, the TPE airport managers should give these attributes the highest priority for improvements.

4.4. Classification of the IPA-Kano model and strategic priorities

Table 7 shows the corresponding category of each service attribute of the TPE airport under the analyses of the IPA, Kano, and IPA-Kano models. In the following section, this study proposes specific strategic actions in terms of different categories of service attributes. From Table 7, the order of strategic priorities to 'Keep up the good work' of the IPA-Kano attribute classification are survival (M), fitness (M), major weapon (O), supportive weapon (O), precious treasure (A), and beginning jewellery (A), and the sequential strategic priorities to improve are fatal (M), chronic disease (M), defenceless strategy point (O),

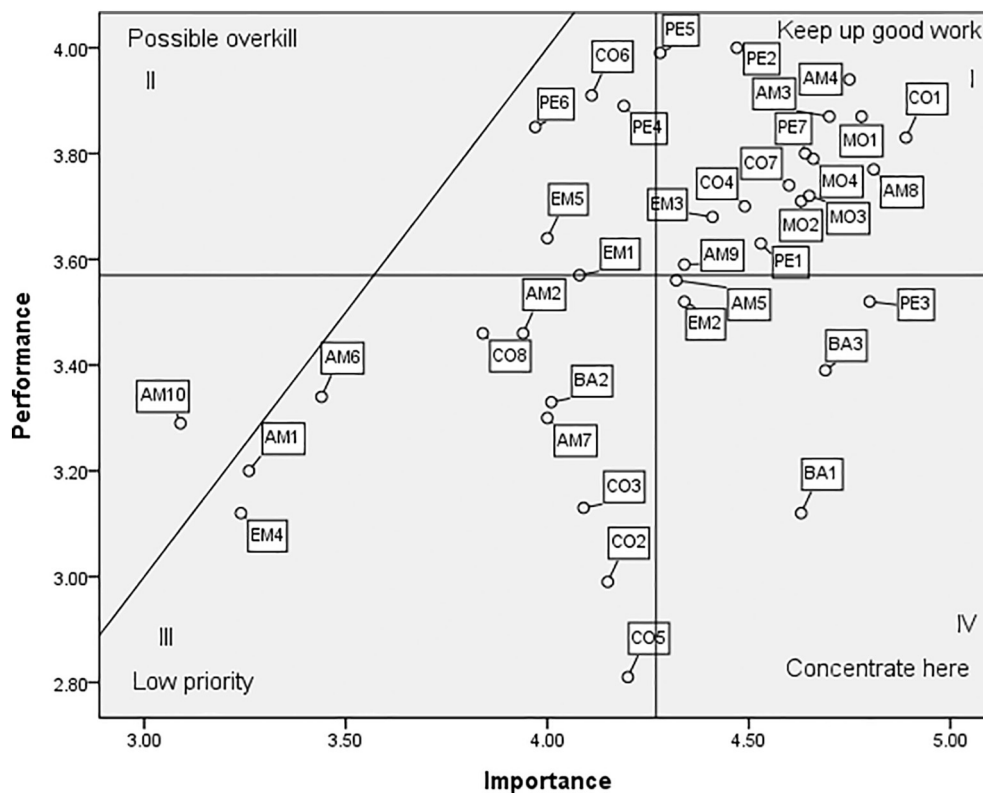


Fig. 1. IPA of TPE airport.

defenceless zone (O), dusty diamond (A), and rough stone (A).

The results of service attribute classification based on the IPA-Kano model show that six attributes, including AM8, PE1, MO1, MO2, MO3, and MO4, are placed in the *survival* attribute category with the first priority to keep up the good work; CO6 and EM5 are placed in the *fitness* category with the second priority; AM3, AM4, CO1, CO4, CO7, PE2, PE5, and PE7 are placed in the *fitness* category with the third priority; PE4 and PE6 are placed in the *supportive weapon* category with the fourth priority; AM9 and EM3 are placed in the *precious treasure* category with the fifth priority; and finally, none are placed in the *beginning jewellery* category.

In contrast, three attributes consisting of AM5, PE3, and BA1 are placed in the *fatal* attribute category with the first priority to improve; AM2, CO3, CO5, EM1, and EM4 are placed in the *chronic disease* category with the second priority; no attribute is placed in the *defenceless strategic point* category with the third priority; only one attribute, CO8, is placed in the *defenceless strategic point* category with the fourth priority; two attributes, EM2 and BA3, are placed in the *dusty diamond* category with the fifth priority; and finally, AM9 and EM3 are placed in the *precious treasure* category with the fifth priority.

5. Discussion

Customer satisfaction is an ever-increasing issue of importance but is not easily addressed. Researchers and practitioners have tried to develop different approaches to address this issue; however, the effectiveness of these proposed methods is in doubt. An excellent and effective method to resolve this difficult problem is not complicated. The introduced IPA and Kano model is the one the managers truly need. The IPA-Kano model is an integrated model that simultaneously considers quality requirement characteristics and the perception and expectation of passengers for each service attribute. The IPA and Kano model analyses each offer their own results and enable development of a corresponding management strategy; however, their respective analysis results do not seem to reach the core problem. The IPA-Kano model

avoids the problem of not considering the two-dimensional quality characteristics of service attributes when only IPA is used, as well as not taking into account the current status of service quality when only the Kano model is applied (Deng, Kuo, & Chen, 2008). Nearly half (43.2%) of the attributes are in the must-be category, indicating that if the attributes are not present, passengers are dissatisfied. Under the framework of IPA, sixteen attributes, including AM3, AM4, AM8, AM9, CO1, CO4, CO7, PE1, PE2, PE5, PE7, EM3, MO1, MO2, MO3, and MO4, are classified in the 'Keep up the good work' quadrant, reflecting that TPE managers should endeavour to maintain the current quality level of these service attributes. Under the framework of the IPA-Kano model, six out of the above sixteen attributes, including AM8, PE1, MO1, MO2, MO3, and MO4, are in the 'survival' category, indicating that it is necessary to maintain these attributes. Otherwise, these attributes may drop into the 'fatal' category, undermining the competitive advantage of the TPE airport. Eight attributes, including AM3, AM4, CO1, CO4, CO7, PE2, PE5, and PE7, are in the 'major weapons' category. The TPE manager must not only maintain the quality level of these attributes to retain passengers but also try to enhance these attributes to repel rivals who show poor performance in these attributes. Two attributes, AM9 and EM3, are in the 'precious treasure' category. In addition to keeping up this good performance, the TPE manager must also work on promoting these attributes to enable these 'precious treasures' to become familiar to the passengers.

Five attributes, including AM5, PE3, EM2, BA1, and BA3, are in the 'concentrate here' quadrant by using the IPA approach, indicating that the TPE airport should first shift their resources to address the quality improvement of these attributes. However, by using the IPA-Kano model, the more accurate priorities of these attributes can be grasped. For instance, three attributes involving AM5, PE3, and BA1 are in the 'fatal' category and thus should receive the highest improvement priority. Two attributes, EM2 and BA3, are in the 'defenceless strategy point' category, which is the third improvement priority.

Table 7
Results of IPA quadrant and IPA-Kano strategic priority

ID	Attribute	KANO	Importance TTIA (A)	Performance TTIA (B)	IPA quadrant	IPA-Kano category	Strategic priority	
							Improvement	Keep up the good work
AM1	Children's play area and facilities provided	I	3.26(L)	3.27(L)	Low priority	Remove		
AM2	Smoking policy/smoking lounges	M	3.94(L)	3.48(L)	Low priority	Chronic disease	2	
AM3	Washroom and shower facilities in terminal	O	4.70(H)	3.89(H)	Keep up the good work	Major weapon		3
AM4	Cleanliness of washroom facilities	O	4.75(H)	3.92(H)	Keep up the good work	Major weapon		3
AM5	Terminal comfort, ambience and design	M	4.32(H)	3.55(L)	Concentrate here	Fatal	1	
AM6	Location of airline lounges	I	3.44(L)	3.38(L)	Low priority	Remove		
AM7	Quiet areas, day rooms, hotel facility, rest areas	A	4.00(L)	3.34(L)	Low priority	Rough stone	6	
AM8	Terminal cleanliness, floors, seating and public areas	M	4.81(H)	3.78(H)	Keep up the good work	Survival		1
AM9	Seating facilities throughout terminals	A	4.34(H)	3.62(H)	Keep up the good work	Precious treasure		5
AM10	TV and entertainment facilities	I	3.09(L)	3.36(L)	Low priority	Remove		
CO1	Getting to and from the airport, ease of access	O	4.89(H)	3.80(H)	Keep up the good work	Major weapon		3
CO2	Choice of bars, cafes and restaurants	A	4.15(L)	3.07(L)	Low priority	Rough stone	6	
CO3	Prices charged in retail outlets	M	4.09(L)	3.21(L)	Low priority	Chronic disease	2	
CO4	Standards of disabled user access and facilities	O	4.49(H)	3.72(H)	Keep up the good work	Major weapon		3
CO5	Prices charged in bars, cafes and restaurants	M	4.20(L)	2.96(L)	Low priority	Chronic disease	2	
CO6	Availability of luggage trolleys (airside and landside)	M	4.11(L)	3.89(H)	Possible overkill	Fitness		2
CO7	Public transport options, efficiency and prices	O	4.60(H)	3.73(H)	Keep up the good work	Major weapon		3
CO8	Choice of shopping - tax free and other outlets	O	3.84(L)	3.53(L)	Low priority	Defenceless zone	4	
PE1	Check-in facilities, queuing systems and seating	M	4.53(H)	3.65(H)	Keep up the good work	Survival		1
PE2	Language skills for airport staff	O	4.47(H)	4.00(H)	Keep up the good work	Major weapon		3
PE3	Immigration - queuing times/system	M	4.80(H)	3.56(L)	Concentrate here	Fatal	1	
PE4	Immigration - staff attitude	O	4.19(L)	3.88(H)	Possible overkill	Supportive weapon		4
PE5	Friendliness of airport staff	O	4.28(H)	3.96(H)	Keep up the good work	Major weapon		3
PE6	Courtesy and attitude of security staff	O	3.97(L)	3.84(H)	Possible overkill	Supportive weapon		4
PE7	Waiting times at security screening	O	4.64(H)	3.80(H)	Keep up the good work	Major weapon		3
EM1	ATM facilities	M	4.08(L)	3.57(L)	Low priority	Chronic disease	2	
EM2	Power charging facilities	A	4.34(H)	3.56(L)	Concentrate here	Dusty diamond	5	
EM3	Wi-Fi service	A	4.41(H)	3.69(H)	Keep up the good work	Precious treasure		5
EM4	Telephone and fax locations	M	3.24(L)	3.22(L)	Low priority	Chronic disease	2	
EM5	Bureau de change facilities	M	4.00(L)	3.66(H)	Possible overkill	Fitness		2
MO1	Way finding and terminal signage	M	4.78(H)	3.84(H)	Keep up the good work	Survival		1
MO 2	Clarity of boarding calls and airport PAs	M	4.63(H)	3.71(H)	Keep up the good work	Survival		1
MO 3	Perception of security and safety standards	M	4.65(H)	3.71(H)	Keep up the good work	Survival		1
MO 4	Ease of transit through airport	M	4.66(H)	3.76(H)	Keep up the good work	Survival		1
BA1	Baggage delivery times	M	4.63(H)	3.21(L)	Concentrate here	Fatal	1	
BA2	Priority baggage delivery efficiency	A	4.01(L)	3.39(L)	Low priority	Rough stone	6	
BA3	Lost luggage services	A	4.69(H)	3.41(L)	Concentrate here	Dusty diamond	5	

6. Conclusion and implications

6.1. Conclusion

The IPA-Kano model incorporating the Skytrax scale is introduced to classify and diagnose the service attributes of the TPE airport. This model uses the most familiar category appellations to accurately depict the implications of the three series and twelve categories of airport service attributes according to customers' perceptions. Consequently, TPE managers can easily, deeply and comprehensively understand the strengths and weaknesses of the essential attributes for the airport and develop effective strategies to address each of them. Moreover, the study also offers two directions to provide TPE managers with strategies to consider how to transfer a service attribute from a category of weakness to one of strength (e.g., from fatal to survival for must-be attributes) as well as how to allocate limited resources to effectively improve passenger satisfaction. Attributes may move from one Kano service category to another and then another as passengers change their attitude about an attribute (e.g., move an attribute from attractive to one-dimensional to must-be), so airport managers should conduct periodic surveys (e.g., seasonally, semi-annually, or annually) to grasp the dynamics of service attributes.

Under the IPA-Kano model, two 'precious treasures' attributes, AM9 and EM3, were found for the TPE airport. This information may provide TPE managers with a hint to create more "precious treasures" attributes, such as improving the performance of their 'dusty diamond' attributes. Additionally, future studies are recommended to include samples from different nationalities to examine whether significant differences exist among different demographic groups in terms of attribute categorization. After service providers apply the introduced IPA-Kano model, the empirical results can offer valuable information for service providers attempting to effectively increase passenger satisfaction and contribute to efforts to reduce the investment of resources on unnecessary attributes.

6.2. Theoretical and managerial implications

The results obtained in this present study provide airport managers with useful and accurate information to improve service quality and heighten passenger satisfaction. The findings present two main theoretical implications. First, the present study applied the IPA-Kano model, which is a three-dimensional framework that simultaneously considers the Kano categorization of an attribute, its importance and its performance. This framework can also be easily employed in different settings for other industries. Second, this research integrates the Kano model and the IPA in identifying which attributes should 'Keep up the good work' and which attributes should be improved, as well as their respective priority regarding upkeep or improvement. This information is valuable for airport managers to effectively allocate limited resources to improve passenger satisfaction. Finally, in addition to the advantages mentioned above, the applied IPA-Kano model is a simple method since it does not require an advanced level of statistical knowledge.

This study also provides three managerial implications. First, as modern medical care pursues precision medicine, service quality management is required to accurately understand and fulfil the needs of customers. Managers have a strong interest in identifying service attributes that are the key drivers of customer satisfaction and dissatisfaction. In this regard, classifying and diagnosing service attributes according to the IPA-Kano model provides managers with more valuable and accurate information to identify the key drivers. Second, the premise for the attractive attributes to be able to exert their effectiveness to greatly increase customer satisfaction is that both 'must-be' and 'one-dimensional' attributes must first reach or surpass a certain satisfaction level with customers. Without that, even though the attractive attribute performed well, it is useless. Finally, extra satisfaction can also be generated by providing the attractive service attributes more quickly

in advance. Airport managers should continuously explore the 'precious treasures' attributes, strongly push 'major weapon' attributes and improve the 'fatal' attributes to create a circumstance of positive, virtuous circles, reinforcing the bond between airport staff and passengers.

This study also has two limitations. The first is that most of the respondents are Taiwanese. Second, the sample is heavily skewed towards younger passengers (age < 40) and females. Thus, the IPA and IPA-Kano strategic results derived by the skewed sample cannot generalize the results to all passengers but are still valuable for the TPE managers. Although the representativeness of sample seems slightly imperfect, it is consistent with the results of the previous works. For example, the survey return rates by females are generally higher than those of males (Curtin, Presser, & Singer, 2000) (Moore & Tarnai, 2002), especially for Internet surveys (Smith, 2008), and survey return rates by young people are generally higher than those of older people (Moore & Tarnai, 2002). Therefore, further study is suggested to obtain more samples from different nationalities.

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